

ZRENJANIN, Serbia

WMO: 131730

Lat: **45.367N**Lon: **20.417E**Elev: **82**StdP: **100.34**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.9	-8.0	-13.2	1.2	-9.7	-10.7	1.5	-7.1	8.9	0.5	8.1	2.4	1.6	290	0.434

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	34.2	21.7	32.5	21.3	30.9	20.7	23.1	30.7	22.2	29.8	21.5	28.8	2.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.9	15.7	26.2	20.0	14.8	25.0	19.1	14.0	24.2	69.0	30.9	65.4	30.0	62.7	28.8	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.5	6.4	5.5	-14.9	36.4	4.9	2.0	-18.4	37.9	-21.3	39.1	-24.0	40.2	-27.5	41.7		
(5)			-15.3	24.4	4.6	1.2	-18.7	25.3	-21.4	26.0	-24.0	26.6	-27.4	27.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.3	0.6	2.7	7.3	12.8	17.5	20.9	22.7	22.9	18.0	12.6	7.4	1.8		
	DBStd	8.98	4.85	5.39	4.79	4.12	3.86	3.73	3.31	3.46	3.90	4.51	4.75	4.68		
	HDD10.0	1018	291	208	110	20	1	0	0	0	0	26	106	257		
	HDD18.3	2663	548	438	343	168	62	17	4	4	53	183	328	514		
	CDD10.0	1868	1	3	26	106	234	328	395	399	240	107	28	2		
	CDD18.3	470	0	0	0	4	37	94	141	145	43	5	0	0		
	CDH23.3	4823	0	0	1	45	349	969	1508	1536	379	37	0	0		
	CDH26.7	1872	0	0	0	6	76	357	641	683	109	2	0	0		
(14) Wind	WSAvg	2.4	2.6	2.8	2.9	2.7	2.3	2.1	2.0	1.8	2.0	2.3	2.7	2.5		
(15) Precipitation	PrecAvg	567	36	37	32	47	58	76	62	48	40	34	46	50		
	PrecMax	681	96	100	100	80	111	140	139	185	83	120	82	128		
	PrecMin	395	3	3	1	17	20	28	13	4	2	0	5	0		
	PrecStd	82	28	28	22	18	27	31	36	40	24	35	19	30		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.2	17.7	21.9	27.4	30.7	34.3	36.0	37.1	32.7	26.3	20.7	14.0		
		MCWB	10.0	10.6	12.9	16.7	19.2	21.9	22.1	21.7	20.1	17.7	14.6	11.0		
	2%	DB	10.7	14.5	19.5	24.4	28.6	32.4	33.9	34.5	29.6	24.3	18.0	11.7		
		MCWB	8.3	9.6	12.0	15.1	18.6	21.5	21.9	21.4	19.0	16.6	13.1	9.3		
	5%	DB	8.9	12.1	17.1	22.2	27.0	30.7	32.2	32.6	27.3	22.1	16.0	9.7		
		MCWB	6.9	8.4	10.9	14.1	18.1	20.9	21.5	21.0	18.3	15.6	12.1	7.7		
	10%	DB	7.2	10.0	14.7	20.2	25.2	28.8	30.3	30.6	25.3	20.0	14.1	7.9		
		MCWB	5.7	7.2	9.6	13.1	17.2	20.1	20.7	20.4	17.6	14.8	11.0	6.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.3	11.4	13.9	17.3	20.7	24.0	24.2	24.4	21.4	18.6	15.4	11.1		
		MCDB	12.8	16.1	20.0	25.0	27.6	30.8	32.3	31.3	29.4	24.8	19.6	13.1		
	2%	WB	8.5	10.0	12.5	15.9	19.6	22.6	23.1	22.9	20.0	17.4	13.6	9.4		
		MCDB	10.4	13.9	18.2	22.7	26.6	29.6	30.7	30.9	27.4	22.7	17.3	11.4		
	5%	WB	7.1	8.7	11.3	14.8	18.7	21.7	22.2	21.9	19.0	16.2	12.5	7.8		
		MCDB	8.6	11.8	16.3	20.8	25.4	28.5	29.7	30.0	25.3	21.0	15.6	9.4		
	10%	WB	5.8	7.4	10.1	13.7	17.8	20.8	21.5	21.1	18.1	15.2	11.2	6.5		
		MCDB	7.2	9.8	14.3	18.9	24.0	27.5	28.7	28.5	23.8	19.5	13.9	7.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	7.2	9.4	10.4	10.9	10.9	11.7	12.2	10.6	9.7	7.0	5.2		
		MCDBR	7.5	10.7	12.6	13.5	13.3	13.5	14.3	14.9	13.6	12.3	9.3	7.2		
	5% WB	MCWBR	5.7	6.7	6.8	6.1	5.3	4.7	4.7	4.8	5.2	6.0	5.5	5.4		
		MCWBR	7.2	9.8	11.4	12.0	12.2	12.0	12.7	12.9	12.0	11.3	9.0	6.6		
(40) Clear-Sky Solar Irradiance	taub	taub	0.346	0.372	0.405	0.461	0.447	0.442	0.449	0.456	0.436	0.399	0.364	0.344		
		taud	2.334	2.260	2.208	2.059	2.166	2.216	2.190	2.171	2.196	2.293	2.345	2.350		
	Ebn at Noon	Ebn at Noon	742	791	816	799	826	831	818	795	775	750	713	701		
		Edh at Noon	80	103	124	156	145	139	141	138	124	99	79	71		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.20	2.04	3.30	4.61	5.60	6.19	6.21	5.52	3.85	2.64	1.58	0.97		
	RadStd	RadStd	0.14	0.28	0.41	0.45	0.51	0.54	0.47	0.52	0.42	0.28	0.13	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.77	N/A	N/A	N/A	+0.55	+0.65	N/A	N/A	N/A	N/A
(47) Regional (52 neighbors)		+0.68	N/A	N/A	N/A	+0.40	N/A	-116	-178	+110	+60

Nomenclature: See separate page